---------- Workshop 5 ---------- (222 Words)

T5.1: Download the 'Portugal_online_retail', 'Sweden_online_retail, and 'UK_online_retail' datasets from Canvas. Apply the apriori algorithm to all datasets using three different confidence levels. Select one confidence level for each dataset that you think works better. Determine the first three most important rules for each dataset using the selected confidence level and report them in the report cell. Explain what each rule means (Completing the report cell is required) (15%).

NOTE: You should comment on your code

In [1]:
```
import pandas as pd
from mlxtend.frequent_patterns import apriori, association_rules
from sklearn.preprocessing import LabelEncoder
import warnings

# Suppressing warnings
warnings.filterwarnings("ignore")

# Defining dataset paths
dataset_paths = {
    'Portugal': "https://raw.githubusercontent.com/maz786/WlvMScAI/main/7
    'Sweden': "https://raw.githubusercontent.com/maz786/WlvMScAI/main/7CS
    'UK': "https://media.githubusercontent.com/media/maz786/WlvMScAI/main
}

def preprocess_dataset(path):
    """Load and preprocess dataset."""
    df = pd.read_csv(path, dtype='unicode')
    df.drop('InvoiceNo', axis=1, inplace=True, errors='ignore')

    # Encoding data categorically
    categorical_cols = df.select_dtypes(include=['object']).columns
    if len(categorical_cols) > 0:
        encoder = LabelEncoder()
        for col in categorical_cols:
            df[col] = encoder.fit_transform(df[col].astype(str))

    df = df.apply(pd.to_numeric, errors='coerce').fillna(0).astype(int)
    return df

def find_best_rules(df, dataset_name):
    """Apply Apriori algorithm with different confidence levels and selec
```

```
    # Min_support adjustment for each dataset accordingly
    min_support = 0.02 if dataset_name == 'UK' else 0.05

    best_average_lift = -1
    best_confidence = None
    best_rules = None

    for confidence in [0.1, 0.2, 0.3]:
        frq_items = apriori(df, min_support=min_support, use_colnames=Tru
        rules = association_rules(frq_items, metric="confidence", min_thr
        rules['length'] = rules['antecedents'].apply(lambda x: len(x))

        # Filtering rules with more than one antecedent for more interest
        rules = rules[rules['length'] > 1]

        if not rules.empty and rules['lift'].mean() > best_average_lift:
            best_average_lift = rules['lift'].mean()
            best_confidence = confidence
            best_rules = rules

    print(f"\nBest confidence for {dataset_name}: {best_confidence}")
    if best_rules is not None:
        best_rules.sort_values('lift', ascending=False, inplace=True)
        return best_rules.head(3)
    else:
        return pd.DataFrame()

# Dataset analysation
for dataset_name, path in dataset_paths.items():
    print(f"\nAnalyzing {dataset_name} dataset...")
    df = preprocess_dataset(path)
    top_rules = find_best_rules(df, dataset_name)
    if not top_rules.empty:
        print(f"Top 3 rules for {dataset_name}:\n{top_rules}")
    else:
        print(f"No significant rules found for {dataset_name}.")
```

```
Analyzing Portugal dataset...

Best confidence for Portugal: 0.1
Top 3 rules for Portugal:
                                                antecedents  \
248739  (LUNCH BAG PINK POLKADOT, CHARLOTTE BAG SUKI D...
290916  (JUMBO BAG VINTAGE LEAF, LUNCH BAG VINTAGE LEA...
290921  (JUMBO BAG VINTAGE LEAF, LUNCH BAG VINTAGE LEA...

                                                consequents  antecedent supp
ort  \
248739  (JUMBO BAG SCANDINAVIAN BLUE PAISLEY, JUMBO  B...            0.051
724
290916  (LUNCH BAG CARS BLUE, LUNCH BAG SPACEBOY DESIG...            0.051
724
290921  (LUNCH BAG CARS BLUE, LUNCH BAG SPACEBOY DESIG...            0.051
724

        consequent support   support  confidence       lift  leverage  \
248739            0.051724  0.051724         1.0  19.333333  0.049049
290916            0.051724  0.051724         1.0  19.333333  0.049049
290921            0.051724  0.051724         1.0  19.333333  0.049049

        conviction  zhangs_metric  length
248739         inf            1.0       5
290916         inf            1.0       5
290921         inf            1.0       5

Analyzing Sweden dataset...

Best confidence for Sweden: 0.1
Top 3 rules for Sweden:
                                               antecedents  \
615    (ASSORTED BOTTLE TOP  MAGNETS, 60 CAKE CASES D...
70956  (SET OF 3 CAKE TINS PANTRY DESIGN, PACK OF 72 ...
70949  (SET OF 3 CAKE TINS PANTRY DESIGN, PACK OF 72 ...

                                               consequents  antecedent suppo
rt  \
615                              (36 DOILIES DOLLY GIRL)           0.0555
56
70956  (RETROSPOT TEA SET CERAMIC 11 PC, VINTAGE HEAD...           0.0555
56
70949                          (BAG 250g SWIRLY MARBLES)           0.0555
56

       consequent support   support  confidence  lift  leverage  convictio
n  \
615              0.055556  0.055556         1.0  18.0  0.052469         in
f
70956            0.055556  0.055556         1.0  18.0  0.052469         in
f
70949            0.055556  0.055556         1.0  18.0  0.052469         in
f

       zhangs_metric  length
615              1.0       2
70956            1.0       5
70949            1.0       6
```

```
Analyzing UK dataset...

Best confidence for UK: 0.1
Top 3 rules for UK:
                                            antecedents  \
164  (GREEN REGENCY TEACUP AND SAUCER, ROSES REGENC...
166  (ROSES REGENCY TEACUP AND SAUCER, PINK REGENCY...
165  (GREEN REGENCY TEACUP AND SAUCER, PINK REGENCY...

                          consequents  antecedent support  \
164   (PINK REGENCY TEACUP AND SAUCER)            0.037553
166  (GREEN REGENCY TEACUP AND SAUCER)            0.029249
165  (ROSES REGENCY TEACUP AND SAUCER)            0.030910

     consequent support  support  confidence       lift  leverage  convict
ion  \
164            0.037660  0.02641    0.703281  18.674462  0.024996    3.243
271
166            0.050035  0.02641    0.902930  18.046041  0.024947    9.786
434
165            0.051267  0.02641    0.854419  16.666089  0.024826    6.516
893

     zhangs_metric  length
164       0.983380       2
166       0.973047       2
165       0.969980       2

Best confidence for Sweden: 0.1
Top 3 rules for Sweden:
                                              antecedents  \
615    (36 DOILIES DOLLY GIRL, 60 CAKE CASES DOLLY GI...
70956  (BAG 250g SWIRLY MARBLES, SET OF 3 CAKE TINS P...
70954  (RETROSPOT TEA SET CERAMIC 11 PC, SET OF 3 CAK...

                                              consequents  antecedent suppo
rt  \
615                       (ASSORTED BOTTLE TOP  MAGNETS)            0.0555
56
70956  (VINTAGE HEADS AND TAILS CARD GAME, RETROSPOT ...            0.0555
56
70954                          (BAG 250g SWIRLY MARBLES)            0.0555
56

       consequent support   support  confidence  lift  leverage  convictio
n  \
615              0.055556  0.055556         1.0  18.0  0.052469         in
f
70956            0.055556  0.055556         1.0  18.0  0.052469         in
f
70954            0.055556  0.055556         1.0  18.0  0.052469         in
f

       zhangs_metric  length
615              1.0       2
70956            1.0       5
70954            1.0       6

Analyzing UK dataset...
```

```
Best confidence for UK: 0.1
Top 3 rules for UK:
                                          antecedents  \
166  (GREEN REGENCY TEACUP AND SAUCER, ROSES REGENC...
165  (PINK REGENCY TEACUP AND SAUCER, ROSES REGENCY...
164  (PINK REGENCY TEACUP AND SAUCER, GREEN REGENCY...

                          consequents  antecedent support  \
166   (PINK REGENCY TEACUP AND SAUCER)            0.037553
165  (GREEN REGENCY TEACUP AND SAUCER)            0.029249
164  (ROSES REGENCY TEACUP AND SAUCER)            0.030910

     consequent support  support  confidence       lift  leverage  convict
ion  \
166            0.037660  0.02641    0.703281  18.674462  0.024996    3.243
271
165            0.050035  0.02641    0.902930  18.046041  0.024947    9.786
434
164            0.051267  0.02641    0.854419  16.666089  0.024826    6.516
893

     zhangs_metric  length
166       0.983380       2
165       0.973047       2
164       0.969980       2
```

1. Suppresses Warnings: Code suppresses warnings for clean output.
2. Defines Dataset Paths: Specifies paths for Portugal, Sweden, and UK datasets.
3. Preprocesses Datasets: preprocess_dataset function loads data, removes 'InvoiceNo', encodes categorical data, converts to numeric, and replaces NaNs with 0.
4. Finds Best Rules: Using Apriori method, find_best_rules function identifies frequent itemsets with varying support thresholds and computes association rules at different confidence levels. It refines rules based on antecedents count, selects optimal confidence level, and returns top three rules based on lift.
5. Analyzes Each Dataset: Iteratively preprocesses and analyzes each dataset, displaying top rules based on chosen confidence level. (97 words)

Association rule mining identifies recurring patterns in databases, aiding understanding of client behavior in retail. For instance, in Portuguese data, buyers purchasing "JUMBO BAG RED RETROSPOT" and "JUMBO BAG PINK VINTAGE PAISLEY" are likely to buy "JUMBO BAG SCANDINAVIAN BLUE PAISLEY," useful for product placement. Swedish data shows correlation between "36 DOILIES DOLLY GIRL" and "PACK OF 60 PINK PAISLEY CAKE CASES." UK data indicates preferences for "GREEN REGENCY TEACUP AND

SAUCER" and "ROSES REGENCY TEACUP AND SAUCER," suggesting "PINK REGENCY TEACUP AND SAUCER" follow-up. Metrics like support, confidence, and lift quantify rule strength. Essential sources include Agrawal et al. (1993), Han et al. (2011), and Hastie et al. (2009), offering foundational knowledge. Practical application insights by Hahsler et al. (2005) provide real-world utilization insights. (125 words)

Bibliography

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